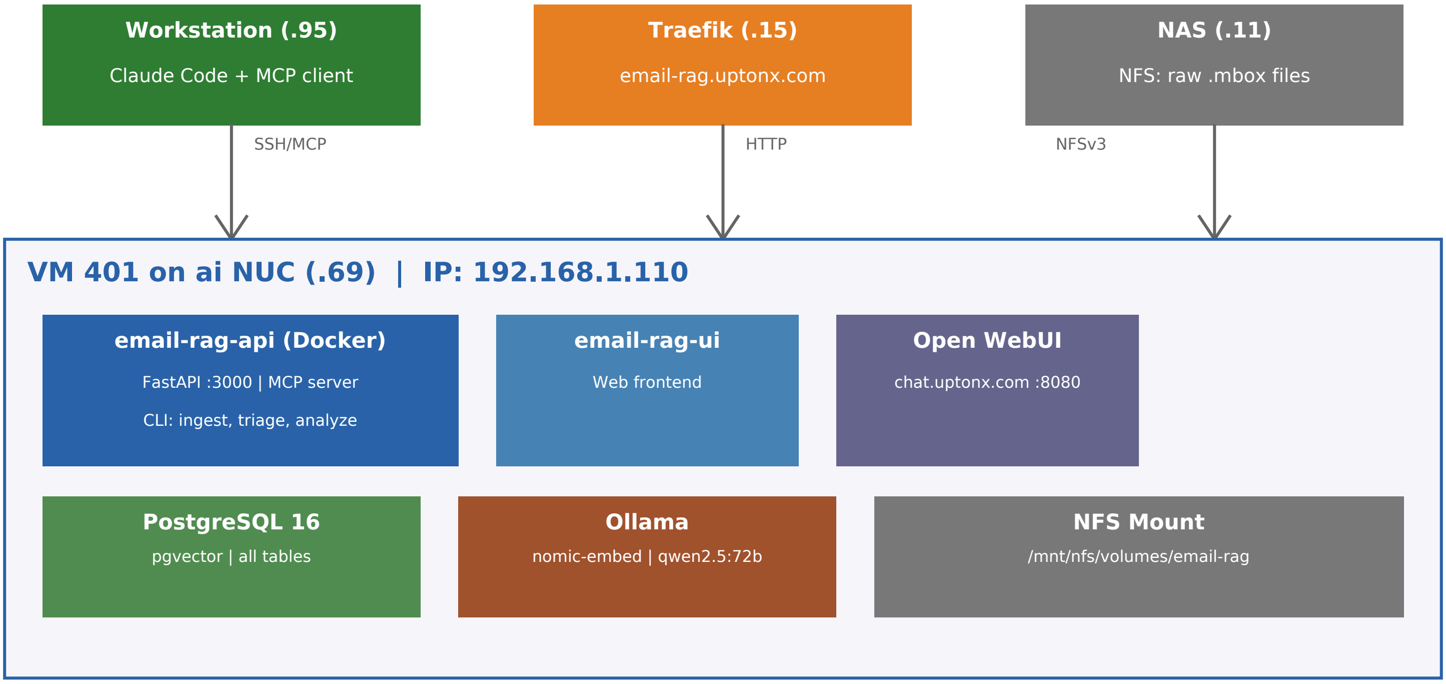
# Email RAG Pipeline

Infrastructure & Architecture Reference
March 2026

## Overview

The Email RAG pipeline ingests Gmail exports (.mbox), stores them in PostgreSQL with pgvector embeddings, and provides semantic search, deep analysis (Ollama LLMs), forensic scanning, and an MCP server for direct Claude Code integration. All components run on a single VM (401) on the ai NUC Proxmox host.

## Infrastructure Diagram




## Repo Structure

```
vm/email-rag/
  docker-compose.yml        # API + UI containers
  Dockerfile                # Python image build
  requirements.txt          # Dependencies (incl. mcp)
  src/email_rag/
    api/main.py             # FastAPI app
    mcp_server.py           # MCP server + anti-speak decoder
    cli.py                  # CLI commands
    db/schema.py            # SQLAlchemy ORM models
    analysis/
      claude_query.py       # Vector search, embeddings
      deep_analysis.py      # qwen2.5:72b analysis
      forensic_decoder.py   # Forensic scanning
```

# Database Schema

| Table | Purpose | Key Fields |
|---|---|---|
| emails | Raw email storage | from, to, subject, body, sent_at |
| snippets | Vector embeddings (pgvector) | content, embedding, email_id |
| findings | Deep analysis results | type, summary, confidence |
| claim_log | Cody's factual claims | claim, source_email_id |
| user_facts | User-verified true/false | fact, verified_status |
| timeline | Real-world events | event_date, description, type |
| anomalies | Forensic scan results | anomaly_type, detail |
| raw_messages | Original MIME content | raw_content, email_id |
| suggested_questions | AI-generated follow-ups | question, context |

# MCP Server Tools

| Tool | Description |
|---|---|
| search_emails | Full-text search across all emails |
| get_thread | Retrieve full email thread by thread_id |
| get_cody_emails | List all emails from Cody |
| semantic_search | Vector similarity search via pgvector |
| decode_text | 3-layer anti-speak decoder |
| get_findings | Retrieve deep analysis findings |
| get_claims | Retrieve claim log entries |
| get_timeline | Retrieve timeline events |
| get_facts | Get user-verified facts |
| add_fact | Store a new verified fact |
| get_anomalies | Retrieve forensic anomalies |
| get_stats | Database statistics overview |

# Network & Access

| Service | Address | Notes |
|---|---|---|
| VM 401 | 192.168.1.110 | ai NUC (.69), user: chris, SSH key auth |
| FastAPI | :3000 | email-rag.uptonx.com via Traefik |
| Open WebUI | :8080 | chat.uptonx.com via Traefik |
| PostgreSQL | :5432 | Local only, pgvector extension |
| Ollama | :11434 | Local, nomic-embed + qwen2.5:72b |
| NFS | .11:/volume1/... | NAS must be on for mbox access |
| MCP | SSH tunnel | ssh chris@.110 docker exec -i |

# Key Stats (March 2026)

Total emails: 706 (Chris + Cody, Jan 2024 - Mar 2026)
Cody's emails: ~310 | Chris's emails: ~396
Verified claims: 32 of 97 | False: 27 (84%) | True: 5
Embeddings: ~611 snippets | Deep analysis findings: ~50+